

PHYSICS-INFORMED LEARNING FOR MATERIALS

From ordinary least squares to PINNs and PINOs

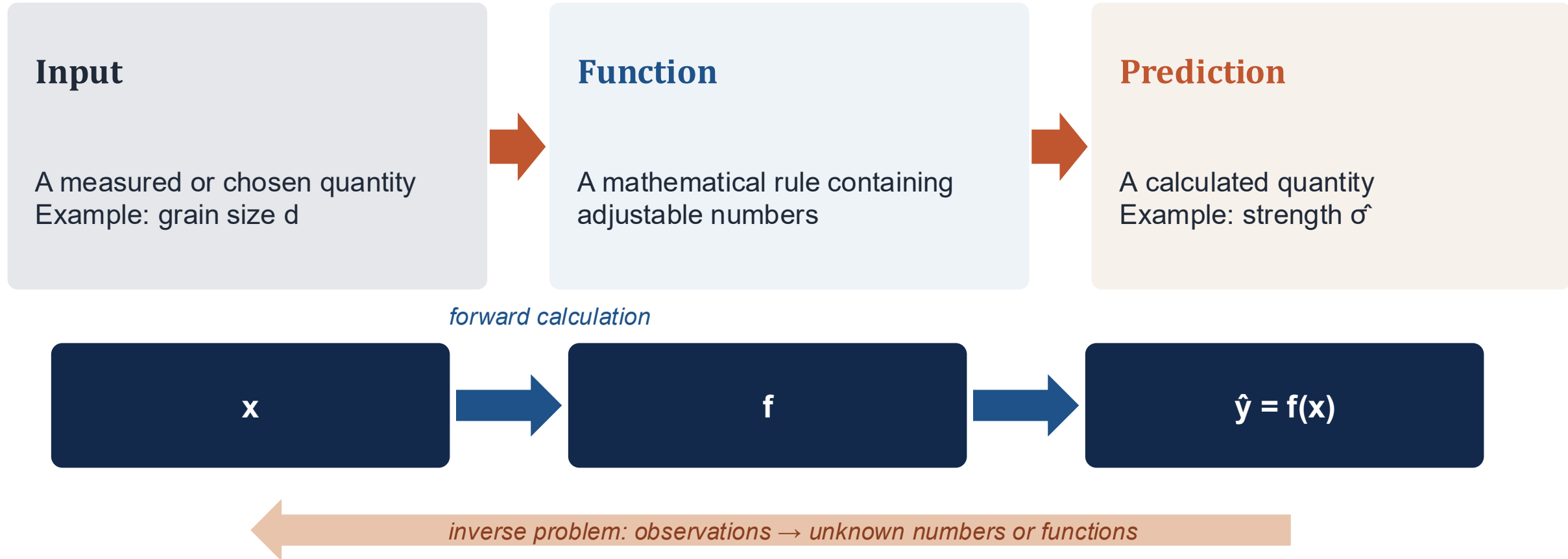
Saswata Bhattacharya

Department of Materials Science and Metallurgical Engineering
Indian Institute of Technology Hyderabad

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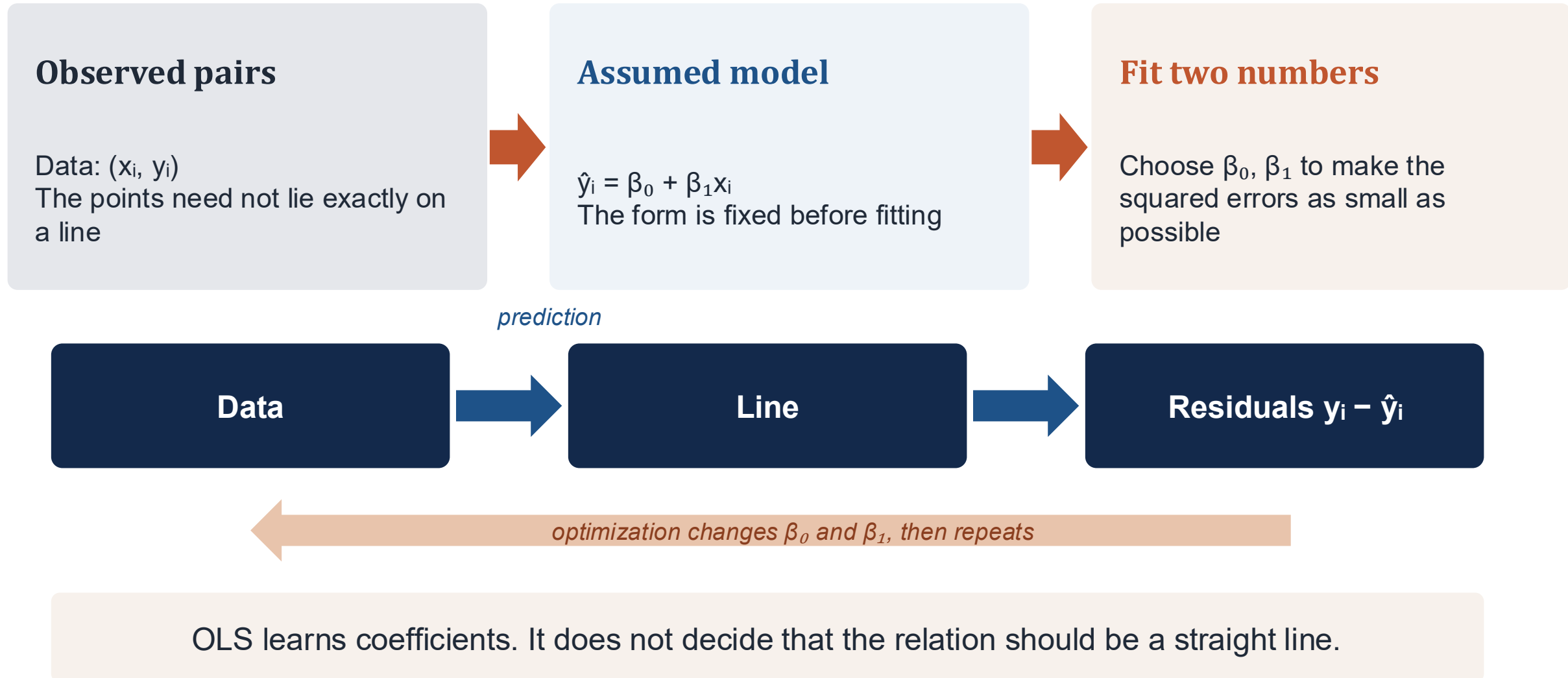


A model maps an input to a prediction

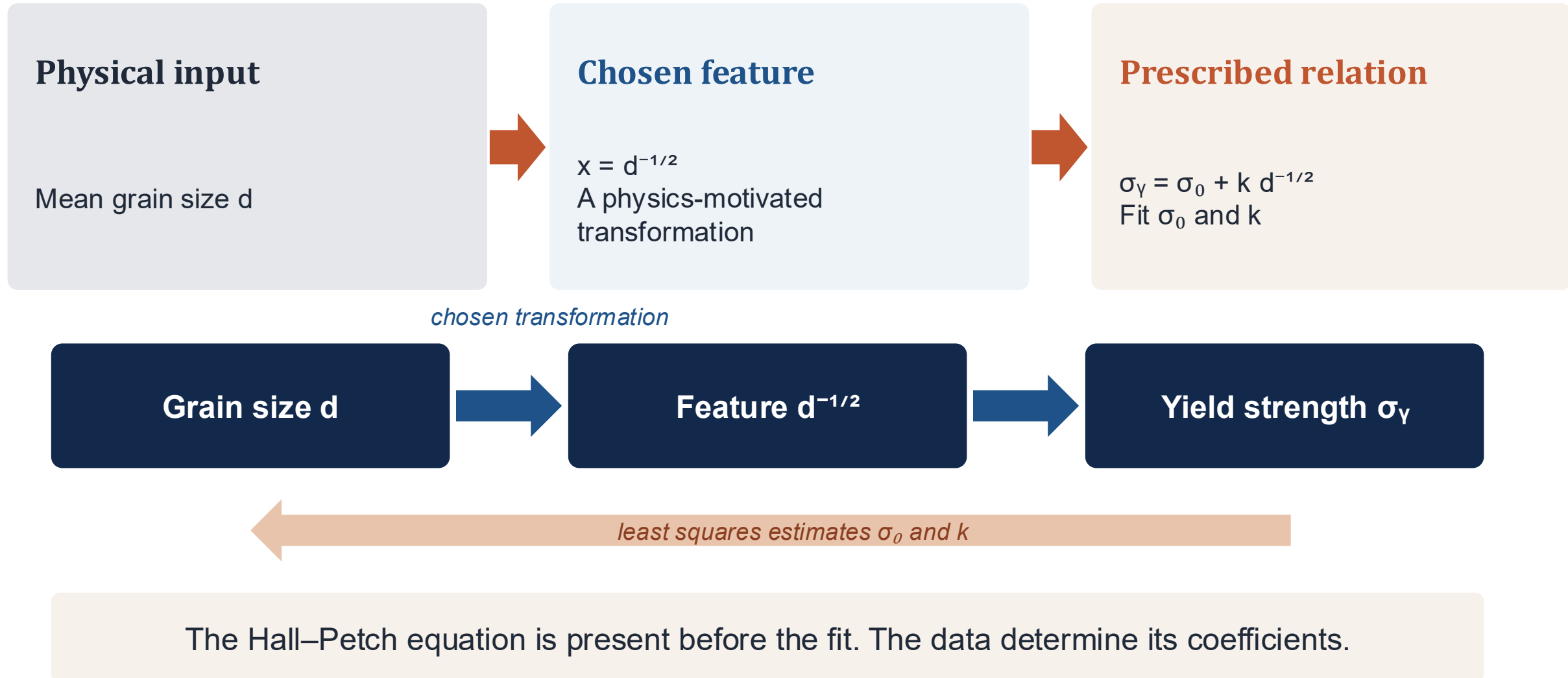


The function will become more flexible, but input → function → output never changes.

OLS begins with a mathematical form that we choose

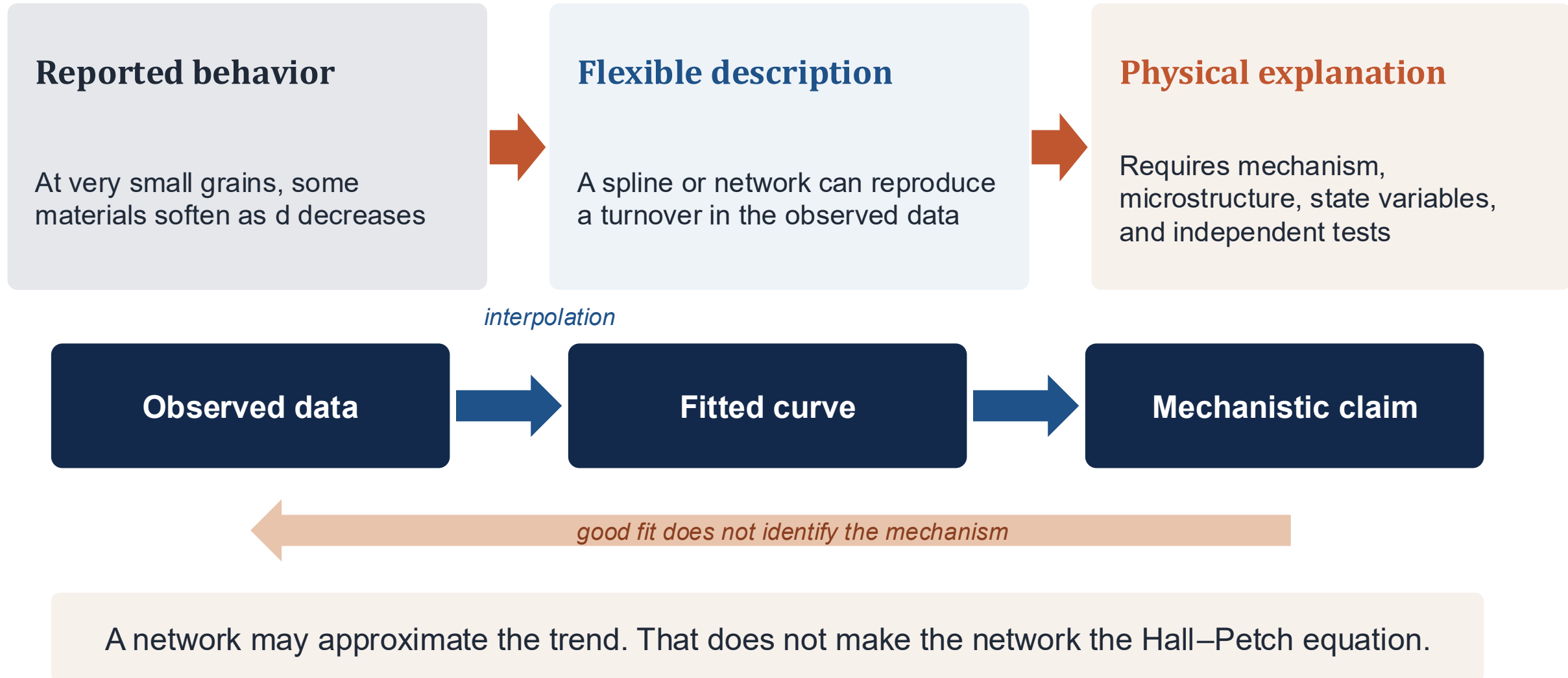


Hall–Petch supplies the form; regression supplies the coefficients

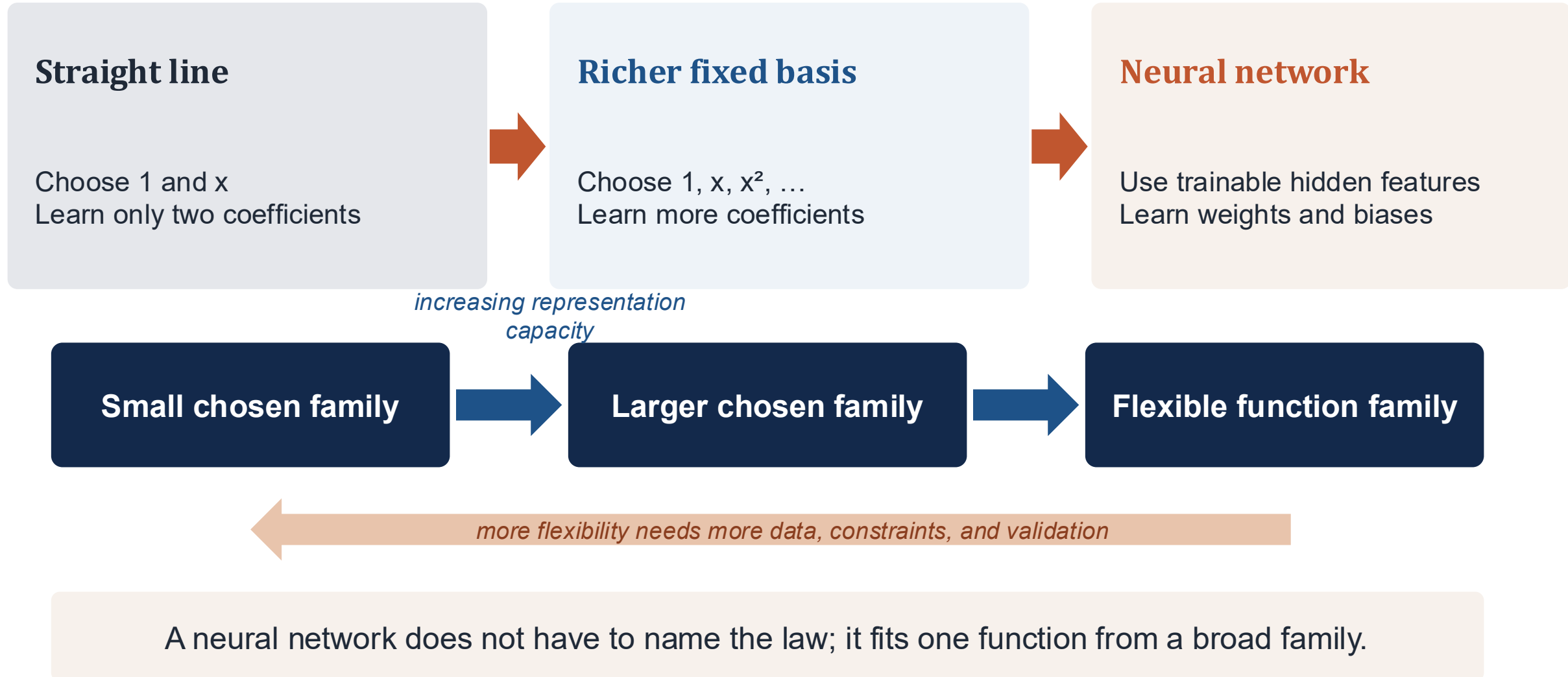


FIT IS NOT MECHANISM

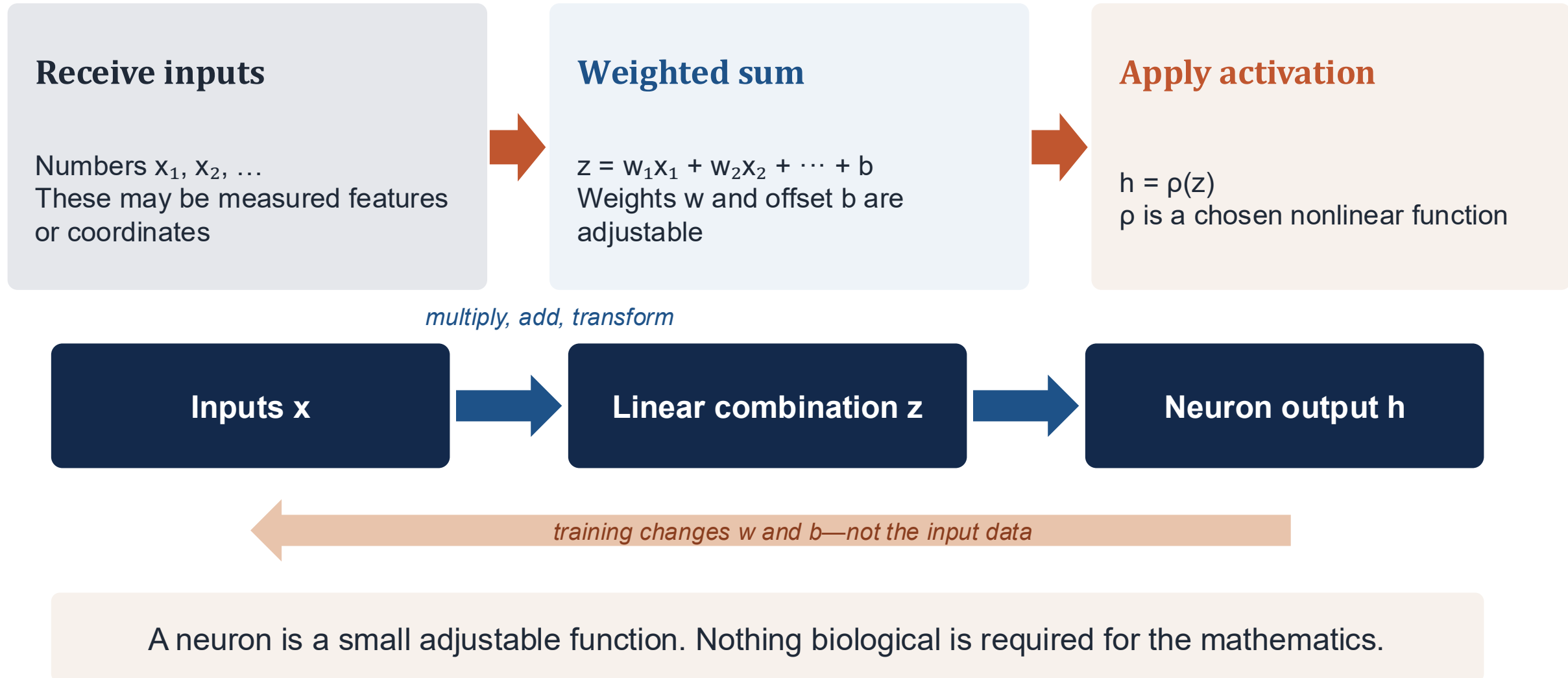
“Inverse Hall–Petch” is a regime—not one universal inverse equation



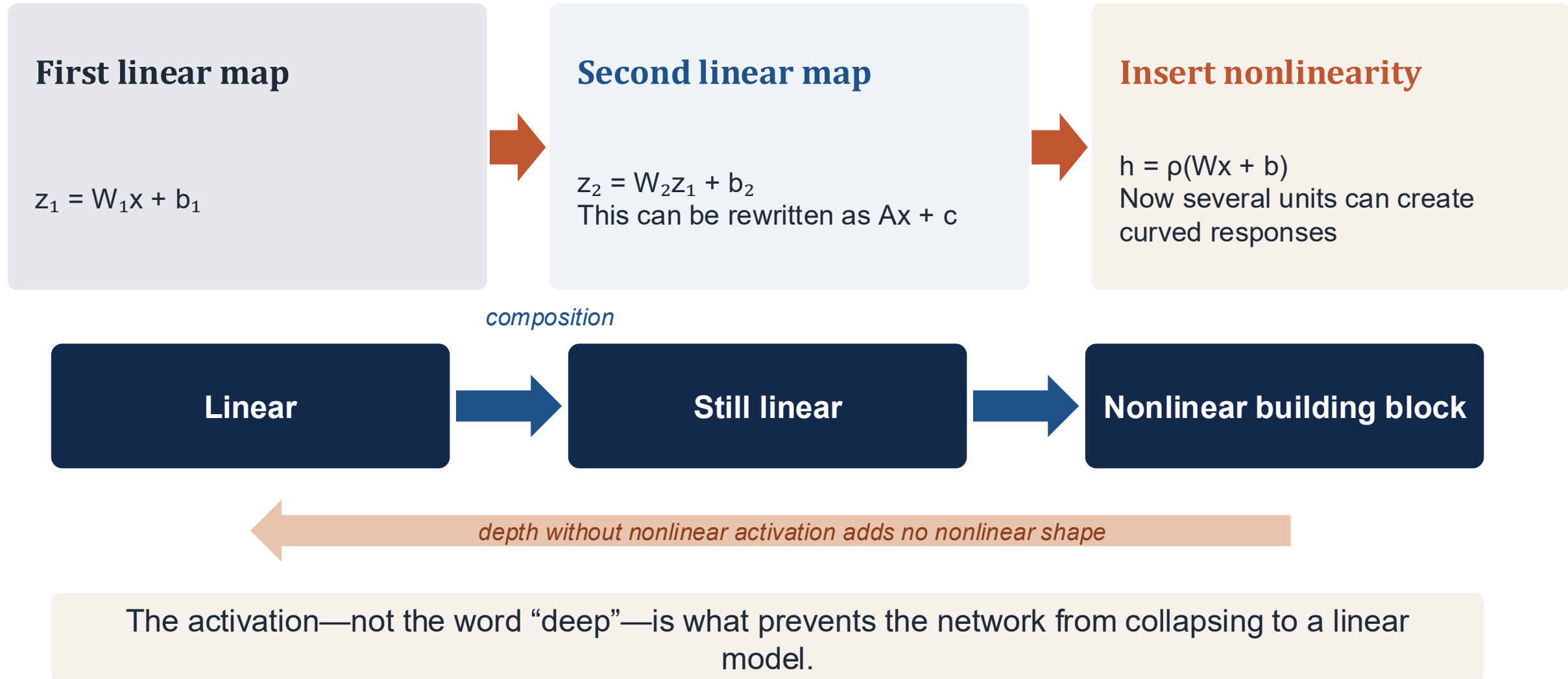
When the shape is unknown, enlarge the family of possible functions



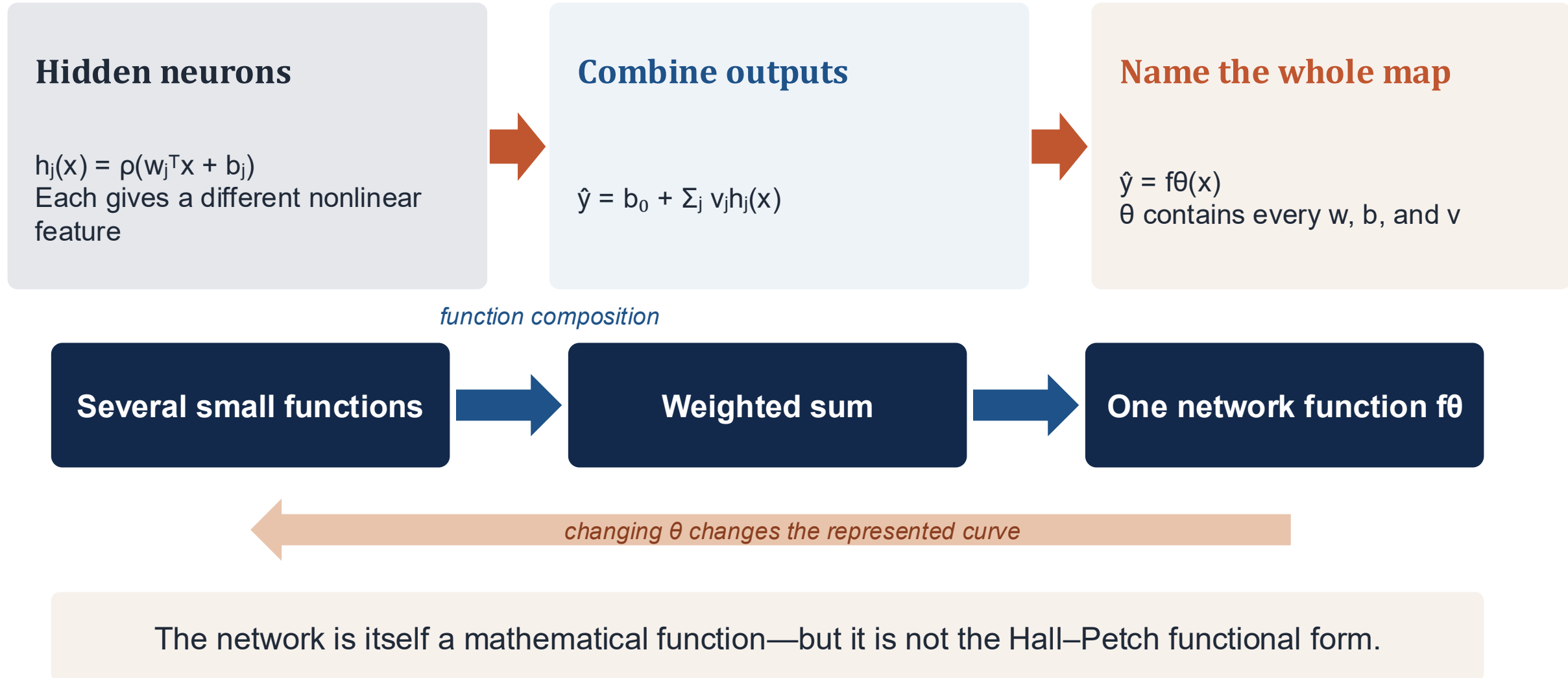
One neuron performs only two elementary operations



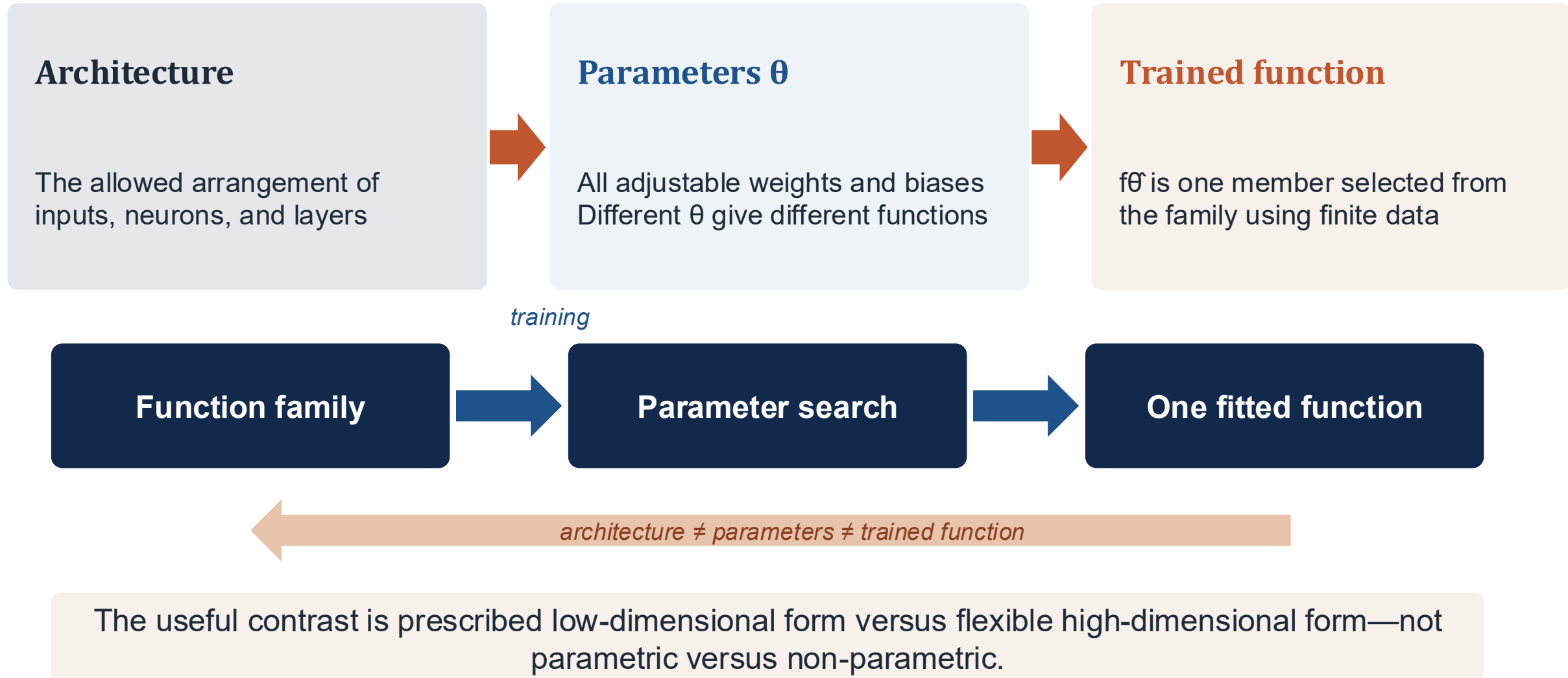
Stacked linear layers are still only one linear map



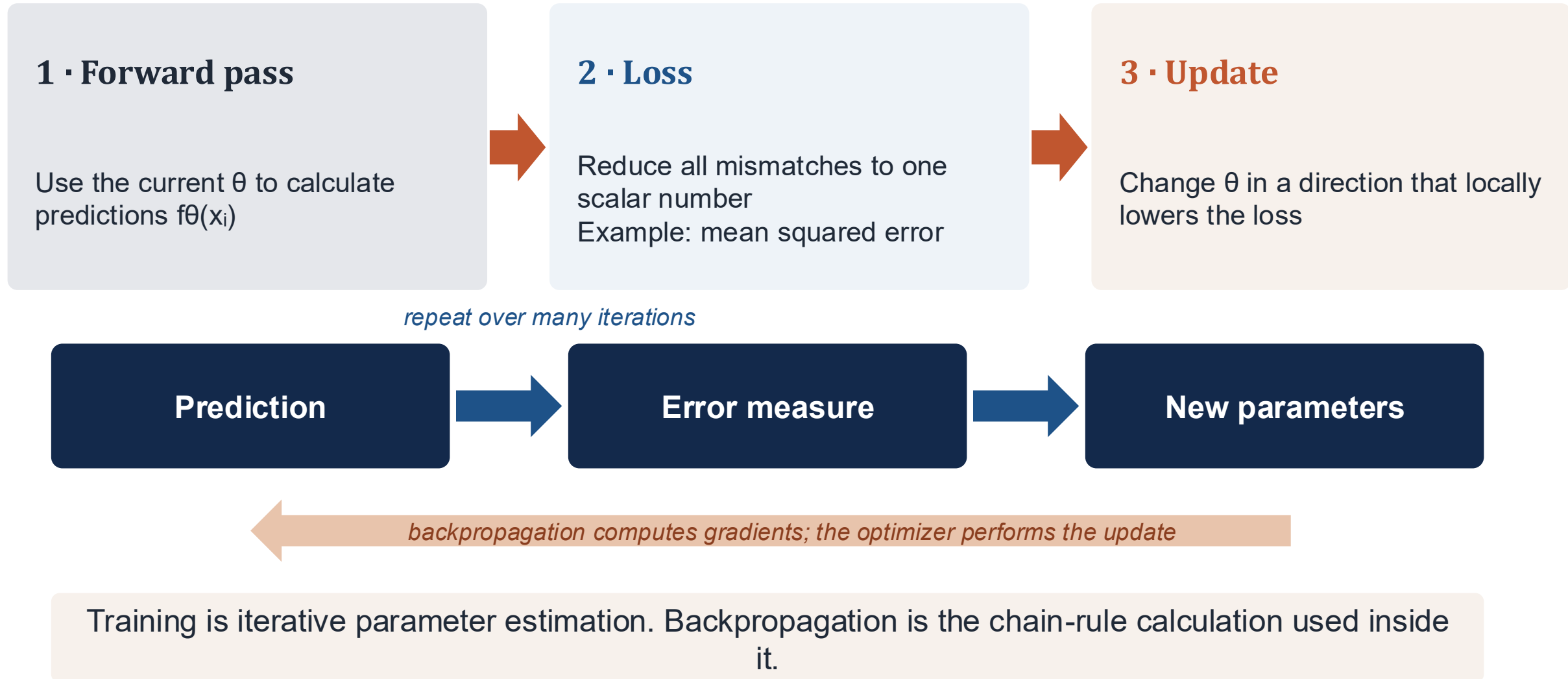
A one-hidden-layer network is one explicit nested function



Both regression models and neural networks have parameters



Training repeats prediction, measurement, and update



Universal approximation describes capacity—not guaranteed learning

The theorem says

A sufficiently rich network family can approximate continuous functions on a bounded domain



It does not say

That training finds the right function, data are sufficient, or extrapolation is correct



Scientific validity needs

Appropriate variables, constraints, independent tests, and uncertainty checks

existence result

Function family



Can approximate



Target function



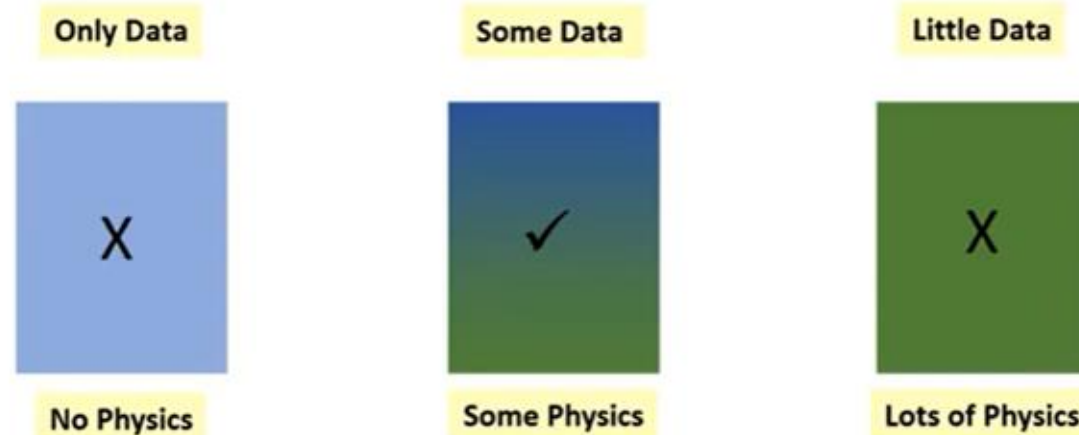
representation ability $\neq$ successful inference

Approximation power does not turn a fitted network into a constitutive law or a mechanism.

FIRST ASK: WHAT SHOULD THE NETWORK REPRESENT?

A network may represent a value, one field, or a map between fields

Three scenarios



The architecture is chosen to match the mathematical object required by the problem

PINN (physics-informed neural network)

Represents one solution field $u(x,t)$. Training checks data, equations, initial conditions, and boundary conditions.

PINO (physics-informed neural operator)

Represents a map from problem inputs to solution fields across a defined family of PDE problems.

PINN: one constrained field · PINO: a constrained map between fields

Phase-field models evolve conserved and non-conserved fields

Free energy F : determines the preferred states

$$F = \int_V [f(c, \eta) + \kappa_c |\nabla c|^2 + \kappa_\eta |\nabla \eta|^2] dV$$

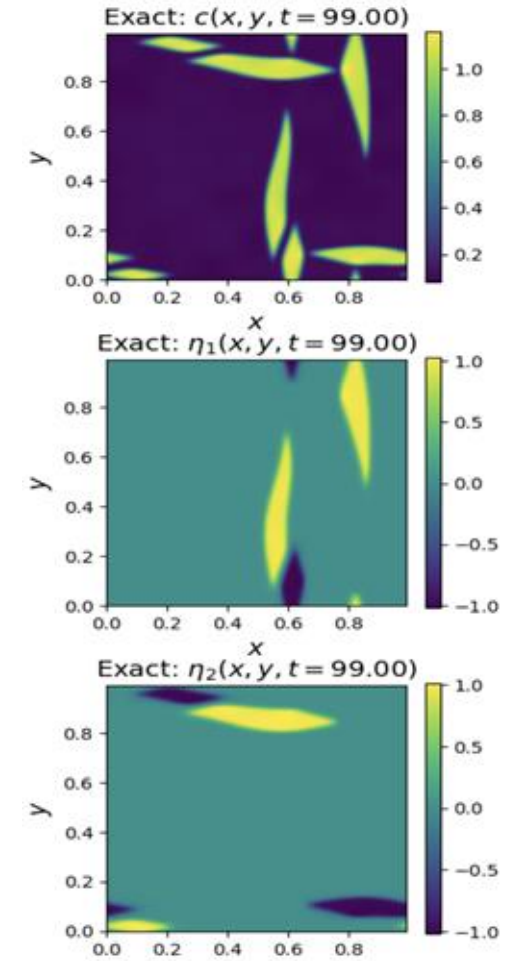
Concentration c : conserved Cahn–Hilliard field

$$\frac{\partial c}{\partial t} = M \nabla^2 \mu, \quad \mu = \frac{\partial f}{\partial c} - 2\kappa_c \nabla^2 c$$

Order parameter η : non-conserved Allen–Cahn field

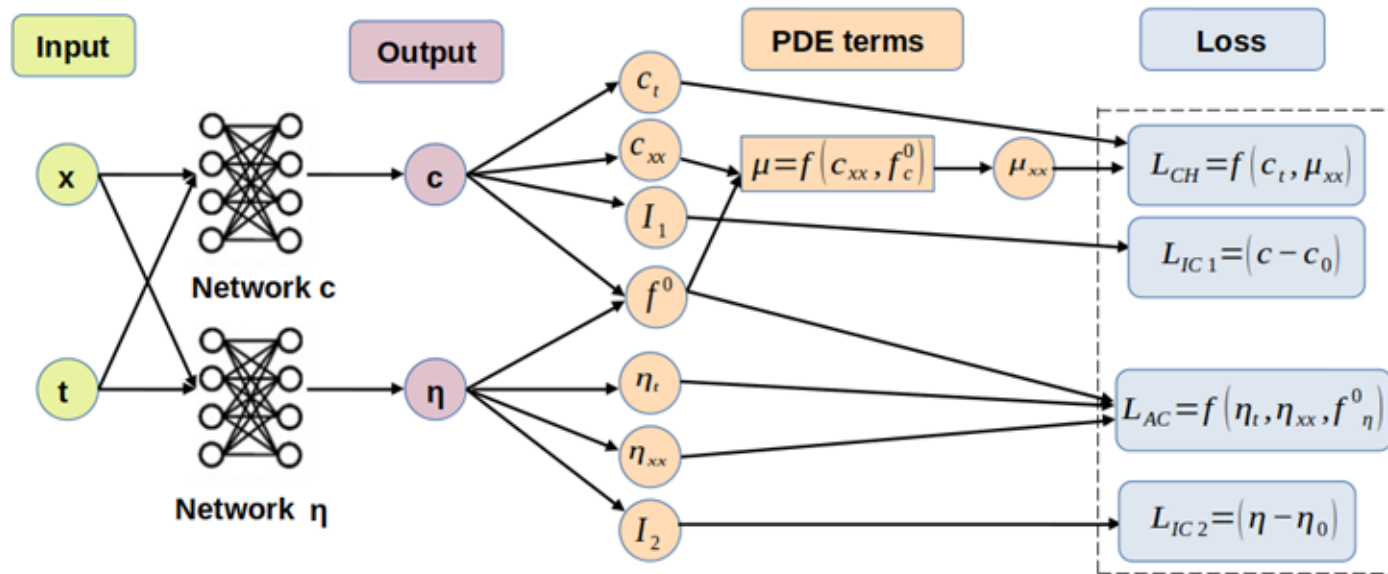
$$\frac{\partial \eta}{\partial t} = -L \left[\frac{\partial f}{\partial \eta} - 2\kappa_\eta \nabla^2 \eta + \frac{\delta F_{el}}{\delta \eta} \right]$$

- Thin interfaces require fine spatial resolution
- Stiff equations require small time steps
- One 4096^2 spectral simulation ≈ 1.8 h



θ' precipitates in Al-Cu : concentration c and order parameters η_1, η_2

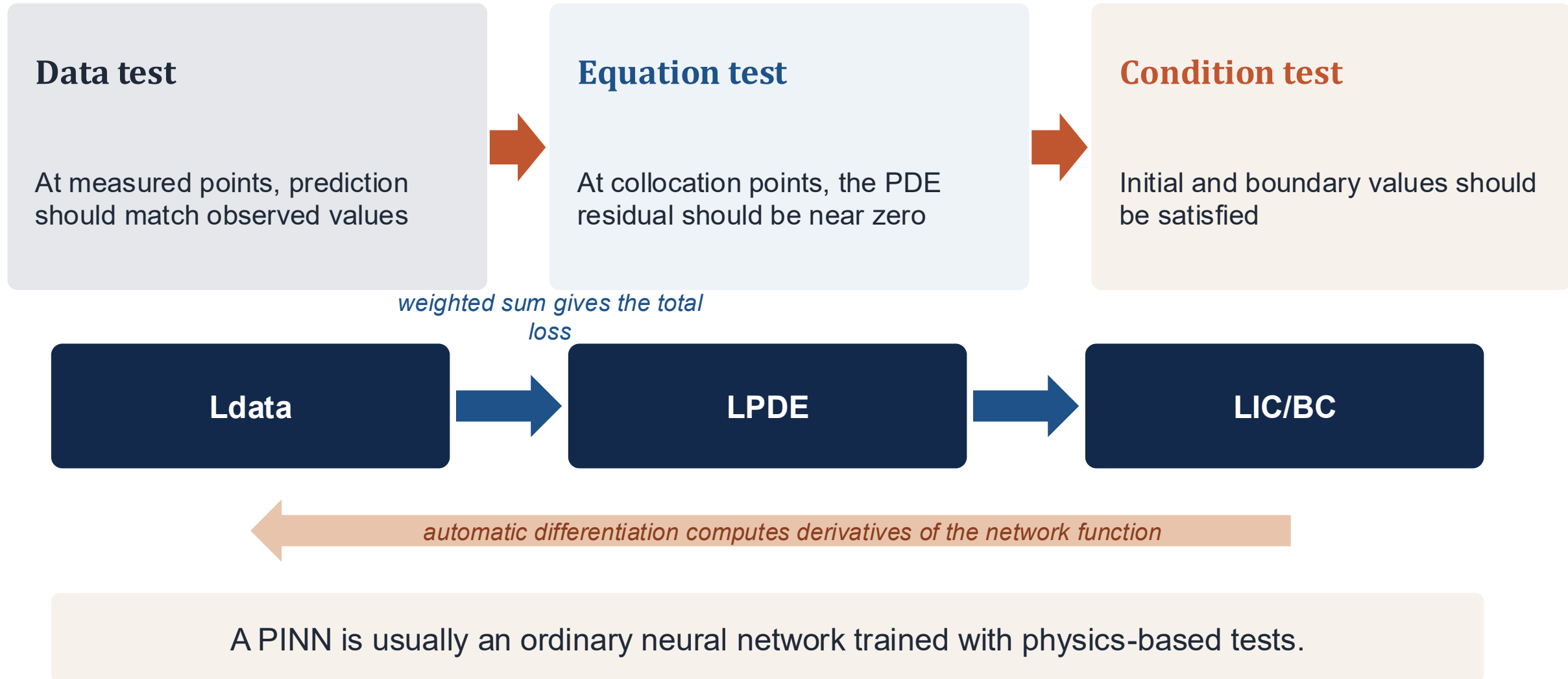
Here the network function represents concentration and order parameters



$$\mathcal{L} = \lambda_1 \mathcal{L}_{CH} + \lambda_2 \mathcal{L}_{AC} + \lambda_3 \mathcal{L}_{IC} + \lambda_4 \mathcal{L}_{data}$$

- Input: position and time
- Output: c and η at that coordinate
- Network function: $[c\theta(x,t), \eta\theta(x,t)]$
- Equation residual: left side – right side
- Automatic differentiation supplies network derivatives
- Training samples points throughout the domain

A PINN loss asks the same network function to pass several tests



Three changes make the optimization manageable

1 · Lower the derivative order

Rewrite one 4th-order equation as two 2nd-order equations

5× faster training

2 · Balance the loss terms

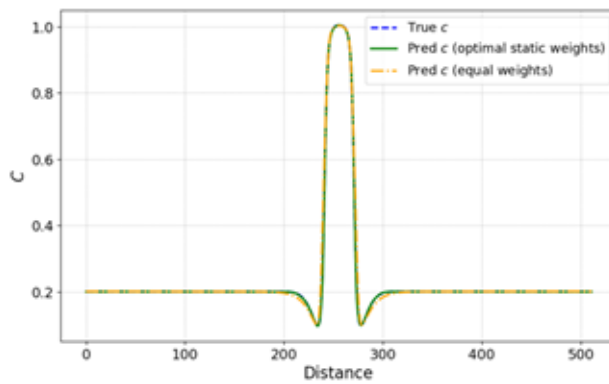
Scale equation and data errors so one term does not dominate

2× lower error

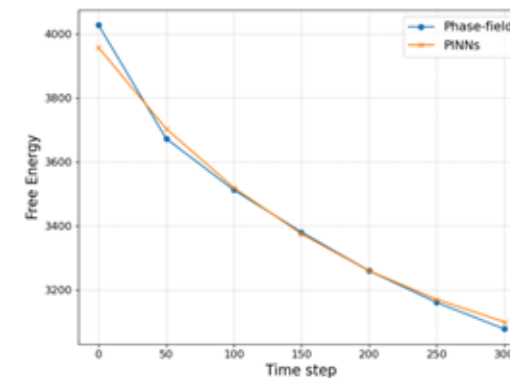
3 · Sample where fields vary

Use more training points near rapidly changing interfaces

Relative field error: $8.6 \rightarrow 7.2 \times 10^{-2}$



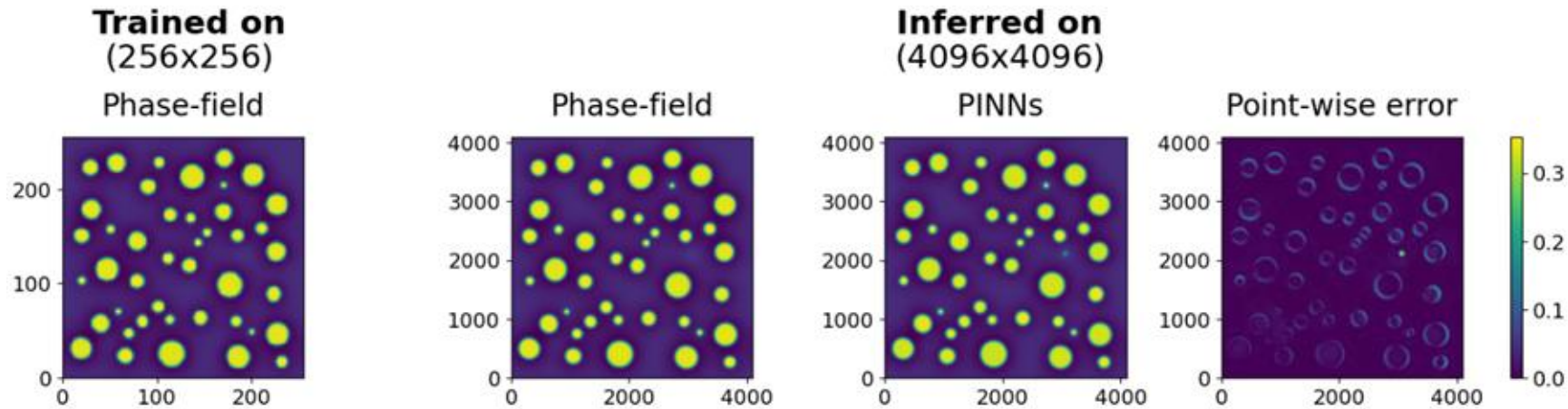
Interface profile: network prediction vs reference



Free-energy decay agrees with the spectral solver

PINN: RESULTS

Train at sampled coordinates, then evaluate the fitted function on a finer grid



16×

finer evaluation grid
 $256^2 \rightarrow 4096^2$

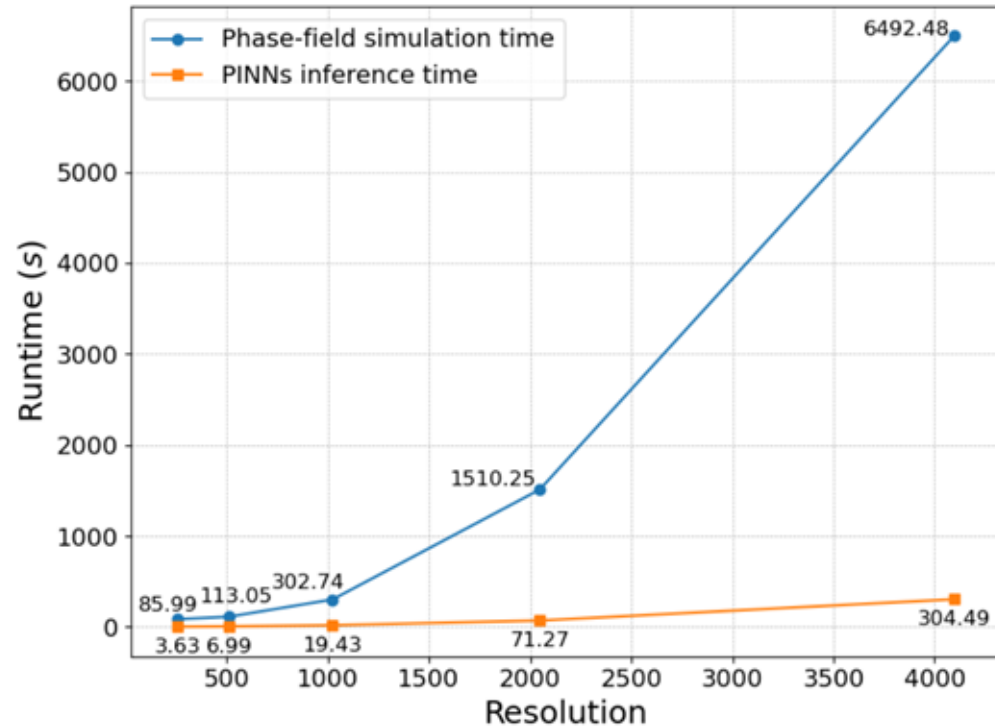
8.8×10^{-3}

mean absolute field error at
 4096^2

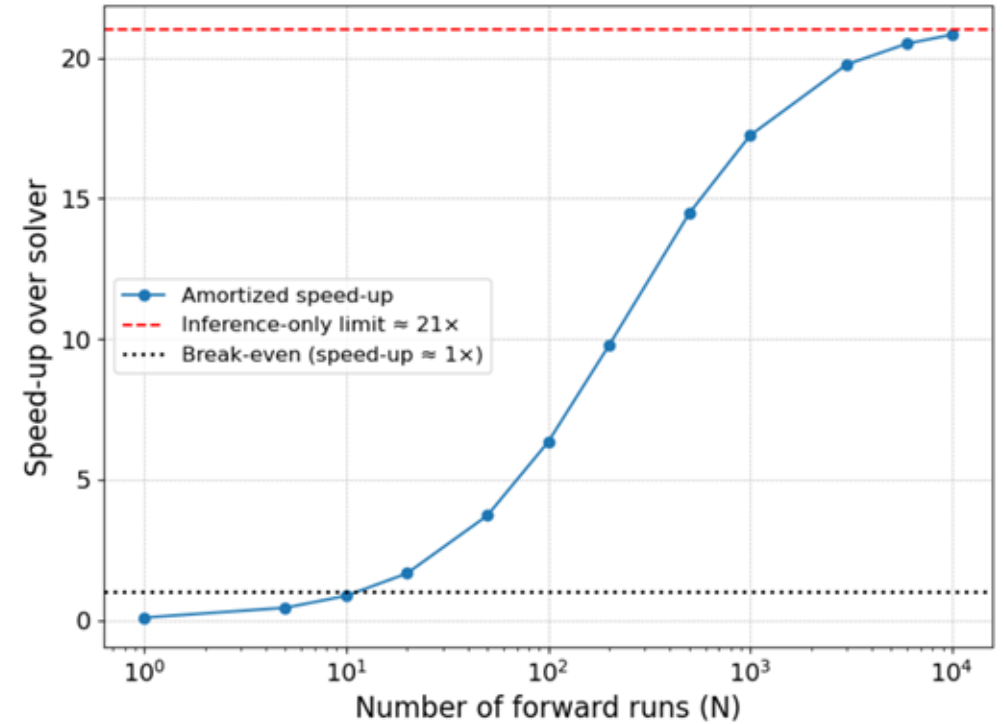
The same fitted function is evaluated at 4096^2 points; it is not retrained

Agreement is checked beyond images: two-point correlation and precipitate volume fraction match the reference

Fast evaluation matters only after the training cost is recovered



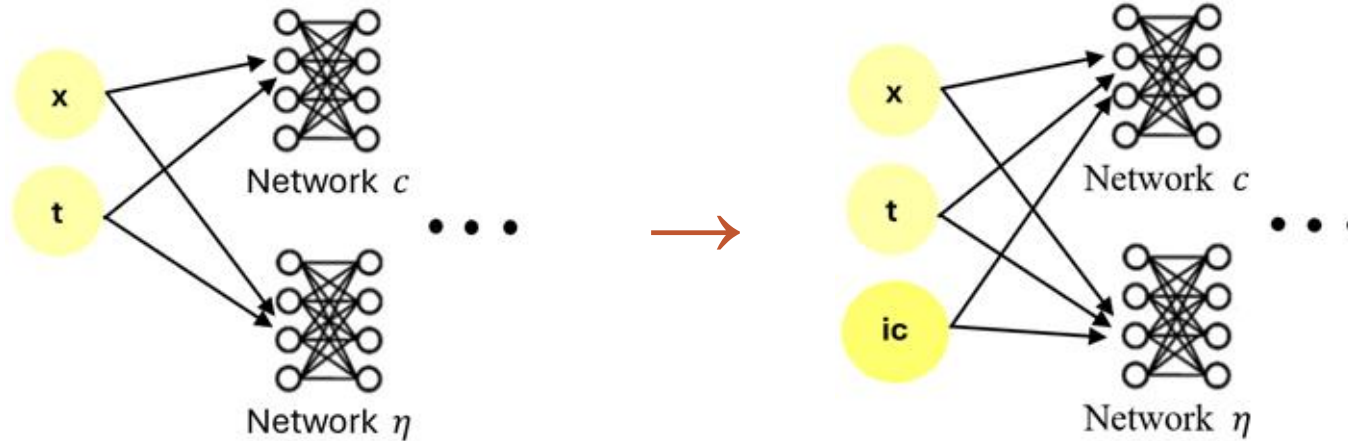
One run: solver 6493 s; trained PINN 304 s at 4096^2



Including training: total cost becomes lower after about 12 runs

21× faster per prediction · break-even after approximately 12 predictions

The initial condition can become an input



Standard PINN: retrain for each initial condition

Conditioned PINN: include the initial field as an additional input

45%

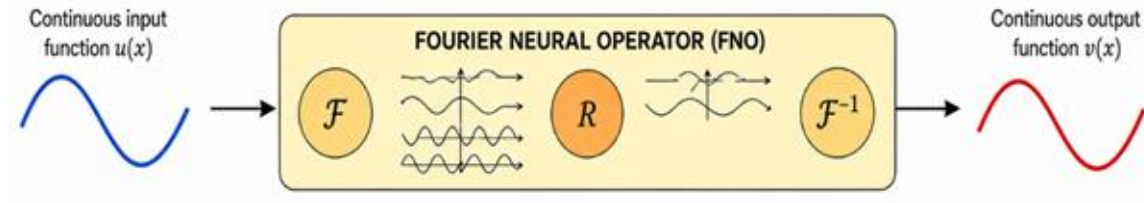
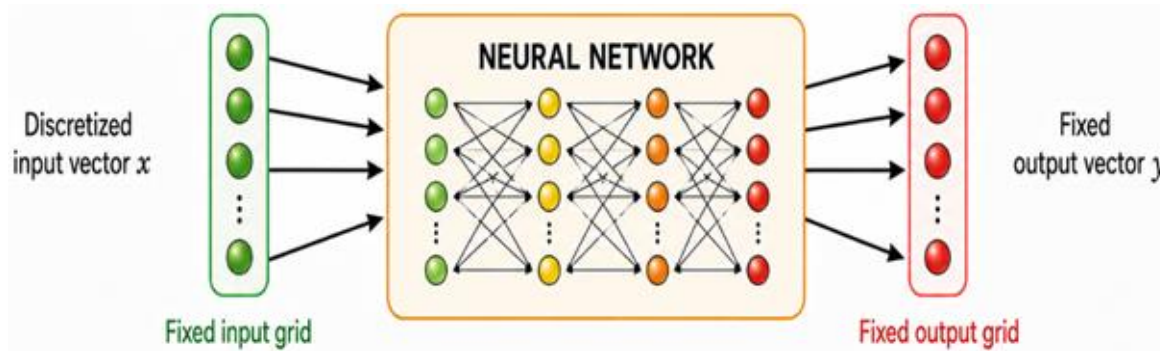
less computation than separate models

$\approx 2\times$

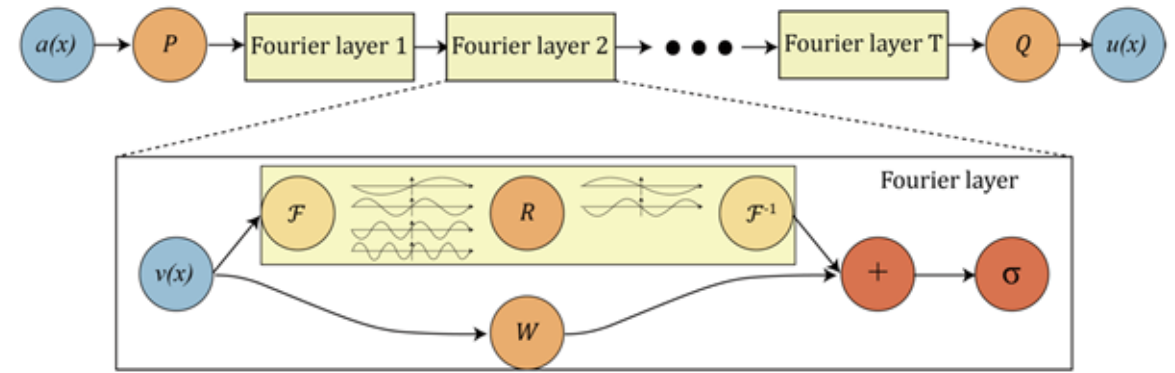
longer time prediction than the training window

This is the bridge to operator learning: the model receives a problem and returns its solution

A neural operator maps an input field to an output field



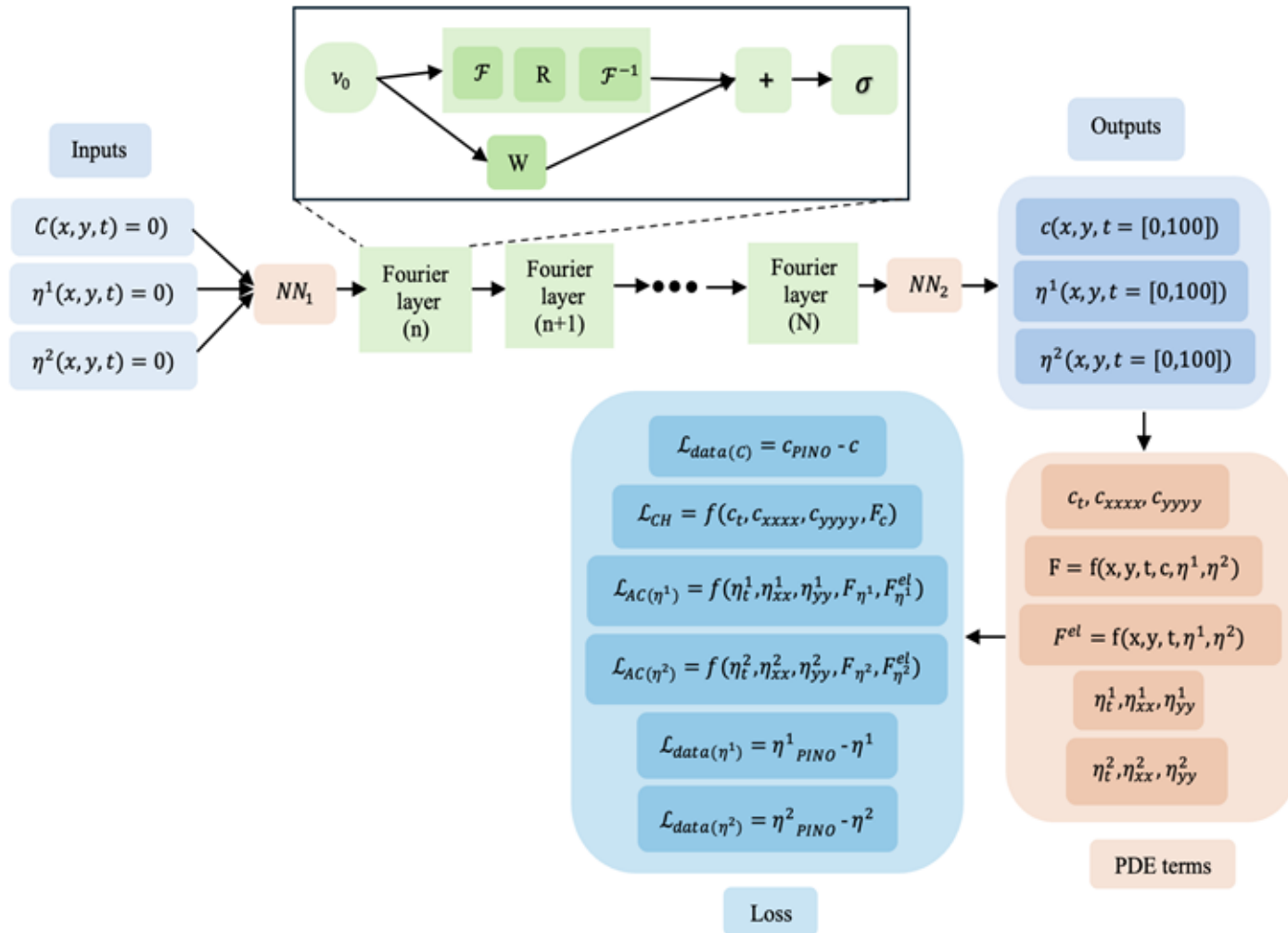
- Ordinary network: finite vector $\rightarrow$ finite vector
- PINN: coordinates $\rightarrow$ values of one solution field
- Neural operator: input field $a(x) \rightarrow$ output field $u(x)$
- Training uses many related problem instances



$$\mathcal{K}(x) = \mathcal{F}^{-1}[R_{\phi} \cdot \mathcal{F}[v_t]](x)$$

- FNO = Fourier neural operator
- 1. Transform the field into Fourier modes
- 2. Learn interactions among retained modes
- 3. Transform back and add a local update
- Compatible finer grids may be queried; accuracy must be checked

PINO predicts a complete precipitate-evolution trajectory



- Input: initial concentration and two order-parameter fields
- Output: their evolution in time
- Training uses data plus residuals of all three equations
- Training set: 12 initial conditions

$$\mathcal{G}_\theta = NN_2 \circ (W_N + \mathcal{K}_N) \circ \dots \circ \sigma(W_1 + \mathcal{K}_1) \circ NN_1$$

Differentiate the represented Fourier field analytically

$$\mathcal{F}\left\{\frac{\partial^4 f}{\partial x^4}\right\} = k^4 \tilde{f}(k)$$

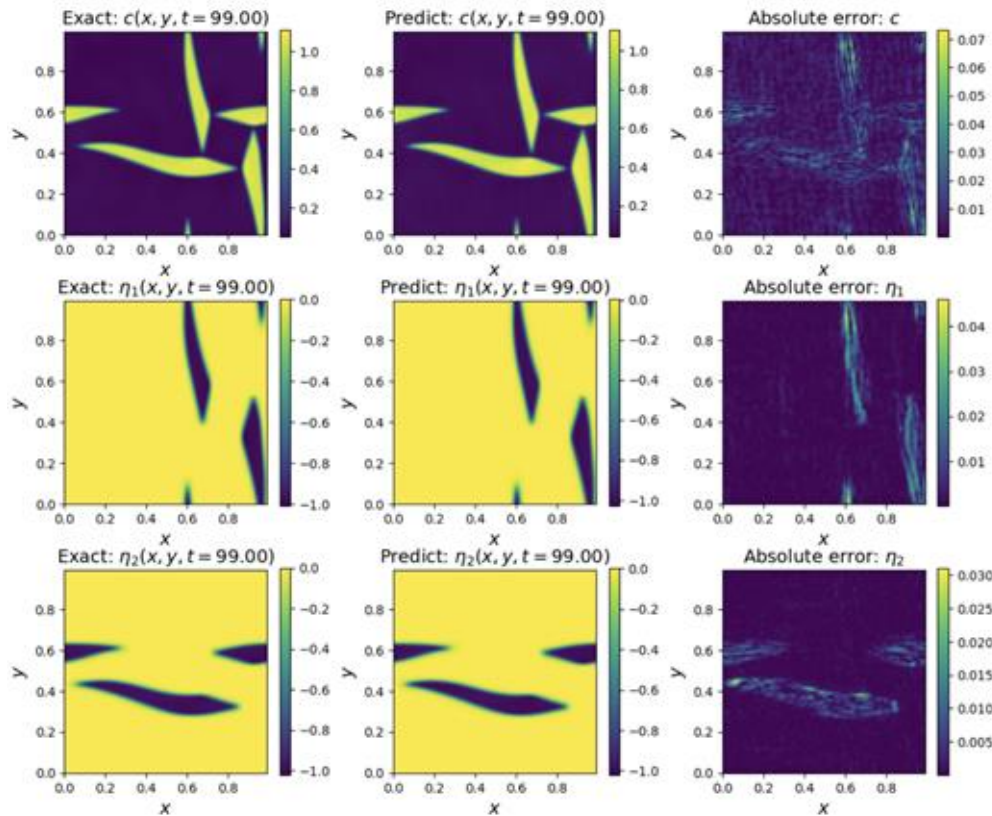
For a retained Fourier mode, a fourth spatial derivative multiplies its coefficient by k^4
This avoids finite-difference stencils

Method	CH residual	Total loss
Finite difference	8.35×10^{10}	1.83×10^{10}
Pseudo-spectral	1.05×10^{-1}	4.03×10^{-2}
Fourier extension	3.31×10^{-1}	9.87×10^{-2}

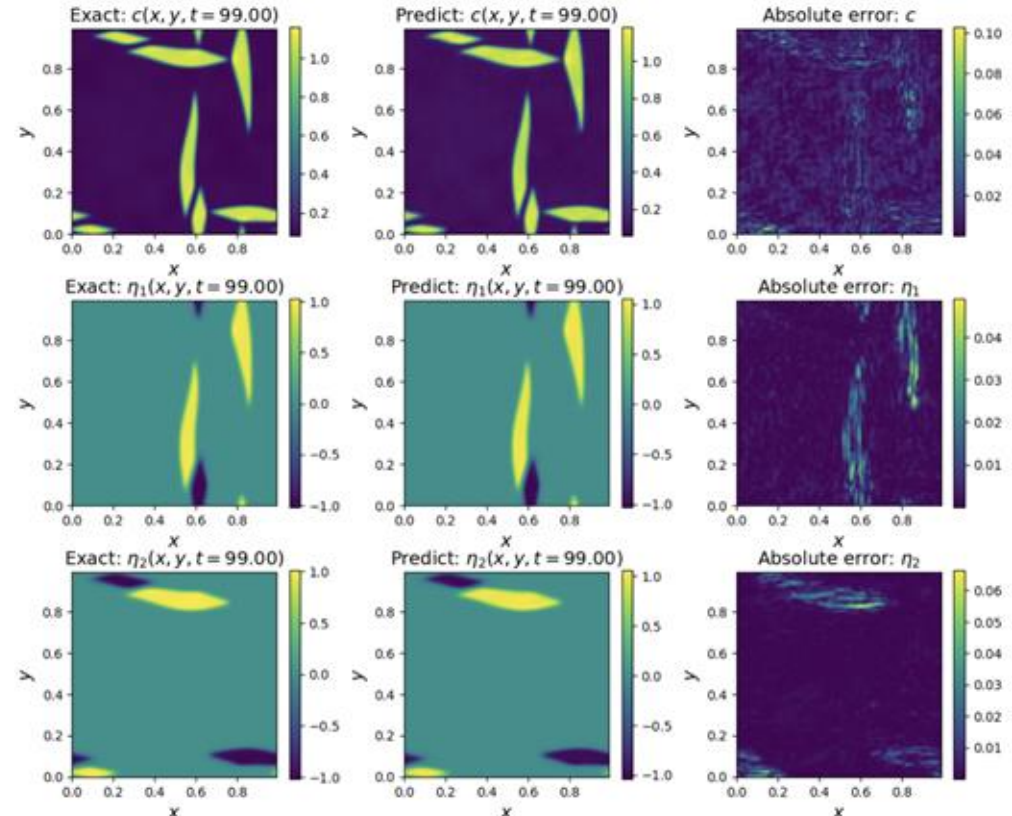
12 orders of magnitude

lower Cahn–Hilliard residual from changing how derivatives of the represented field are computed

Test the trained operator on initial fields not used for training



Unseen within the defined problem family: reference | prediction
| error; relative L_2 error $\approx 2-6 \times 10^{-2}$



Different morphology, same trained operator; no parameter
update

10× faster at 256^2 · evaluated at 256^2 after training with 64^2 data

PART II

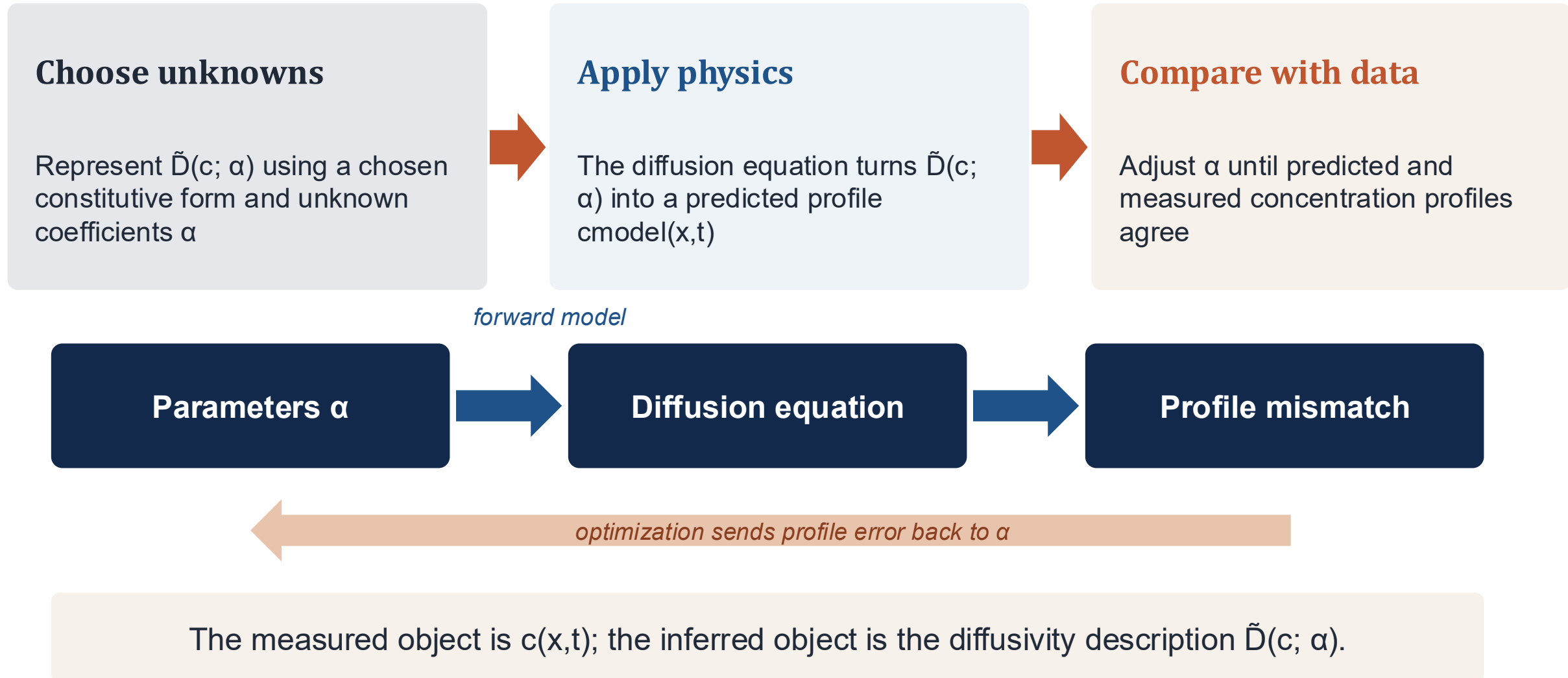
Inverse problems: from measured profiles to diffusion parameters

Forward problem: parameters $\rightarrow$ concentration profiles.

Inverse problem: measured profiles $\rightarrow$ unknown parameters.

We first write the binary version in one line. Then the same logic extends to multicomponent transport, where profiles, thermodynamic coupling, tracer information, and composition-dependent kinetic parameters must be considered together.

A diffusion inverse problem is regression through a differential equation



The Boltzmann variable reduces $c(x,t)$ to one collapsed profile $c(\lambda)$

Profiles in x and t

A diffusion couple provides concentration c at position x and time t

Similarity coordinate

$$\lambda = (x - x_M)/\sqrt{t}$$

x_M is the Matano plane

Collapsed description

Learn $c\theta(\lambda)$
Check the transformed diffusion equation as an ODE residual

Boltzmann transformation

$c(x,t)$

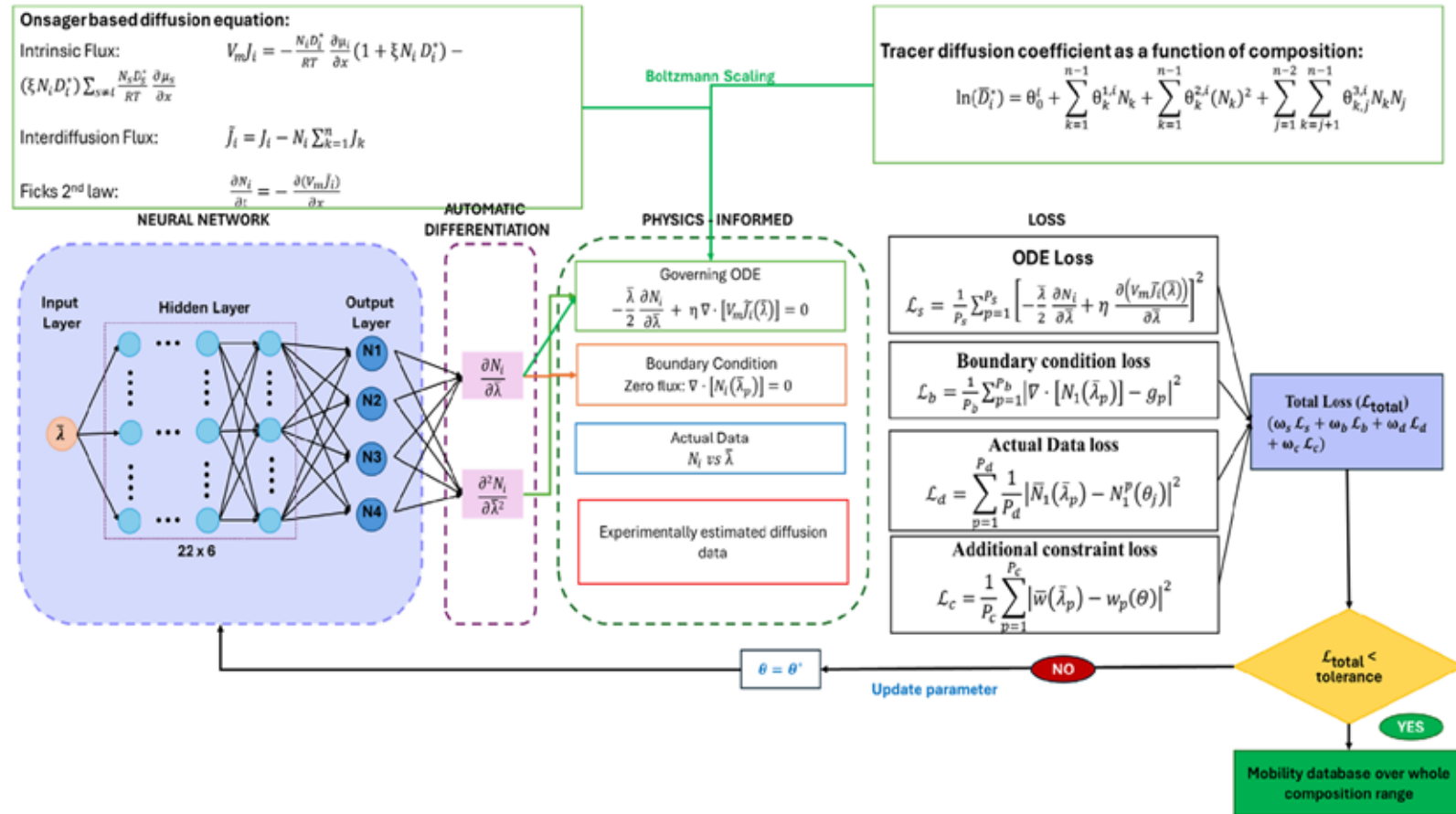
Change coordinate

$c(\lambda)$

one coordinate makes the first PINN example easier to inspect

The network represents $c\theta(\lambda)$. The constitutive diffusivity $\tilde{D}(c; \alpha)$ remains a separate model.

The network profile and the diffusivity model are not the same equation



$$-\frac{\bar{\lambda}}{2} \frac{dN_i}{d\bar{\lambda}} + \eta \frac{d}{d\bar{\lambda}} [V_m \bar{J}_i(\bar{\lambda})] = 0, \quad \lambda = x/\sqrt{t}$$

- Neural network: represents concentration profiles $c_i, \theta(\lambda)$

- Constitutive model: $D_i^*(c; \alpha)$ has a chosen form with unknown coefficients α

- Physics: Onsager transport equations connect profiles and diffusivities

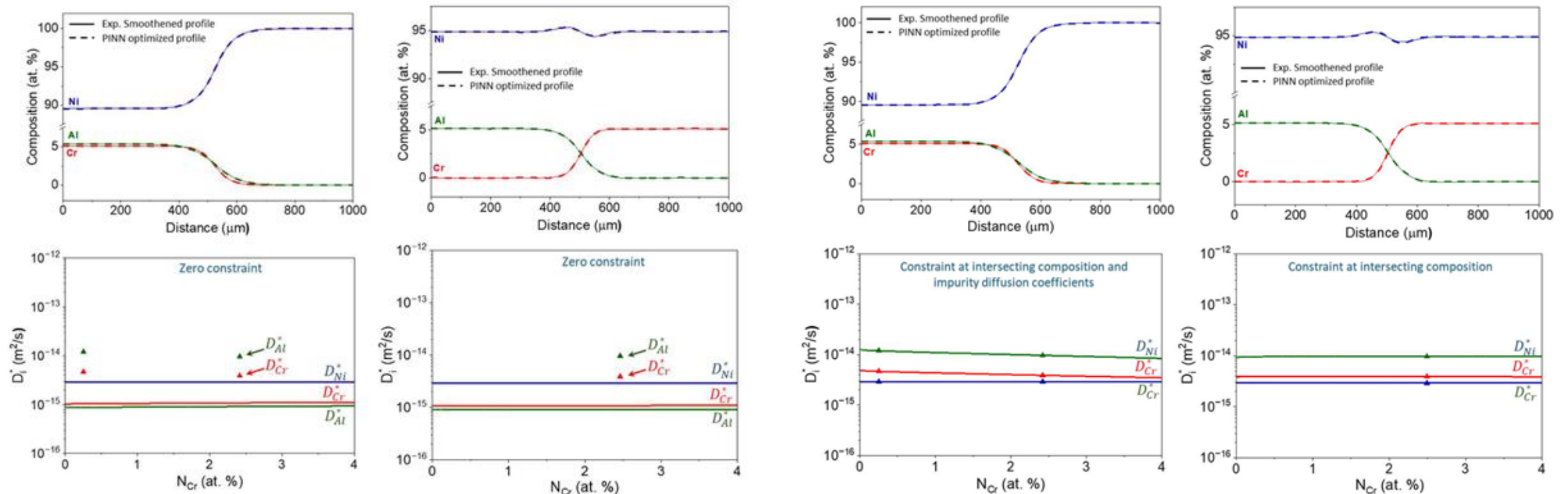
- Training: adjusts θ and α using profile data and equation residuals

- Independent tracer measurements act as anchors for α

$$\mathcal{L}_{total} = \omega_s \mathcal{L}_{ODE} + \omega_b \mathcal{L}_{BC} + \omega_d \mathcal{L}_{data} + \omega_c \mathcal{L}_{constraint}$$

A perfect profile fit can still give the wrong diffusivities

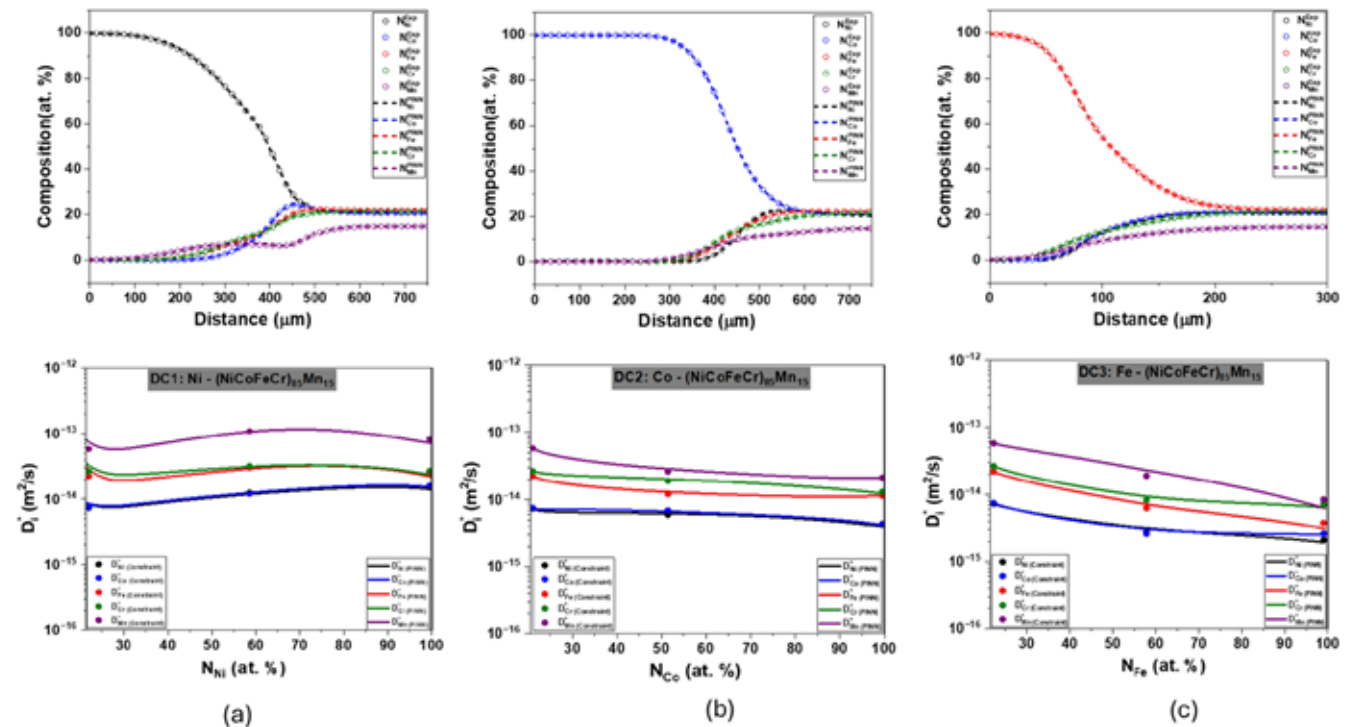
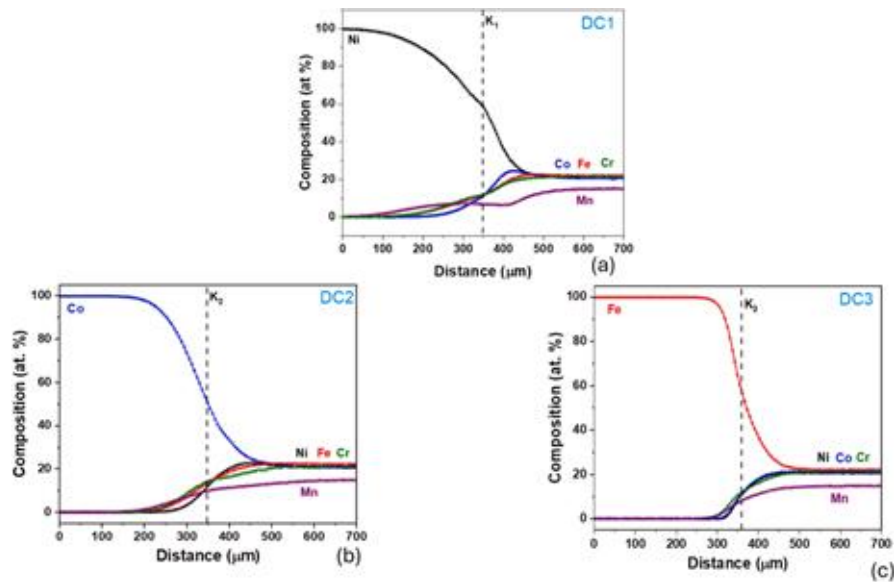
Profile-only inversion is non-unique: several parameter sets can reproduce nearly identical profiles



Profiles only: excellent fit, but wrong tracer-diffusion ordering

Profiles + measured anchors: same profile fit, correct tracer diffusivities

Three diffusion couples constrain 75 mobility parameters

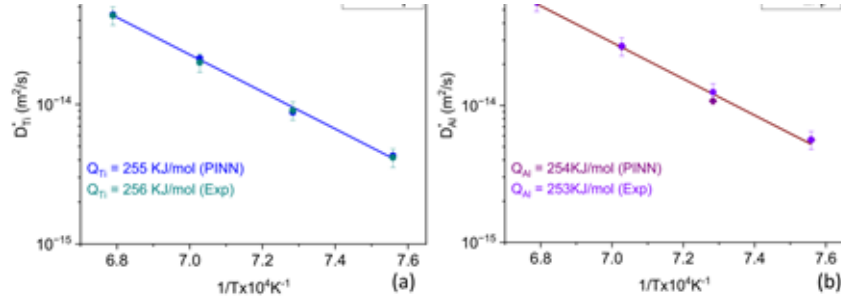


Three designed couples at 1200 °C: Ni, Co, and Fe against $(\text{NiCoFeCr})_{85}\text{Mn}_{15}$

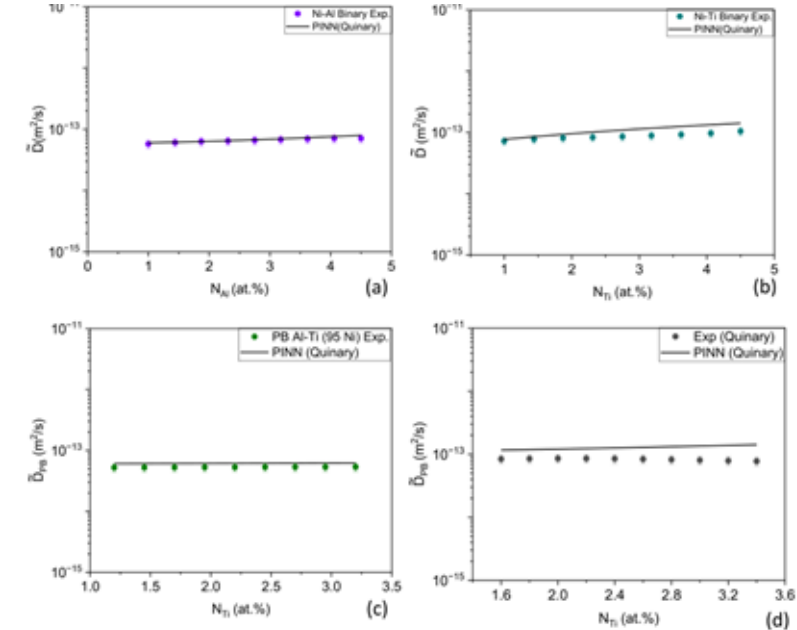
Estimated tracer-diffusion functions for all five elements across the sampled composition range

3 diffusion couples → 75 composition-dependent parameters

Independent experiments test transfer beyond the fitted couples



Arrhenius test: temperature dependence of extracted tracer diffusion in Ni–Al–Ti



Five-component fit also reproduces binary and pseudo-binary observations

255 vs 256

Q(Ti): activation energy, PINN vs experiment (kJ/mol)

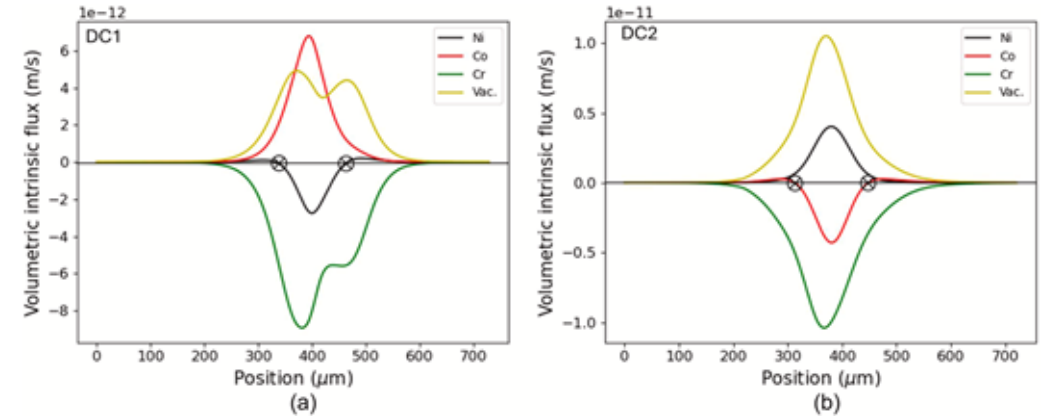
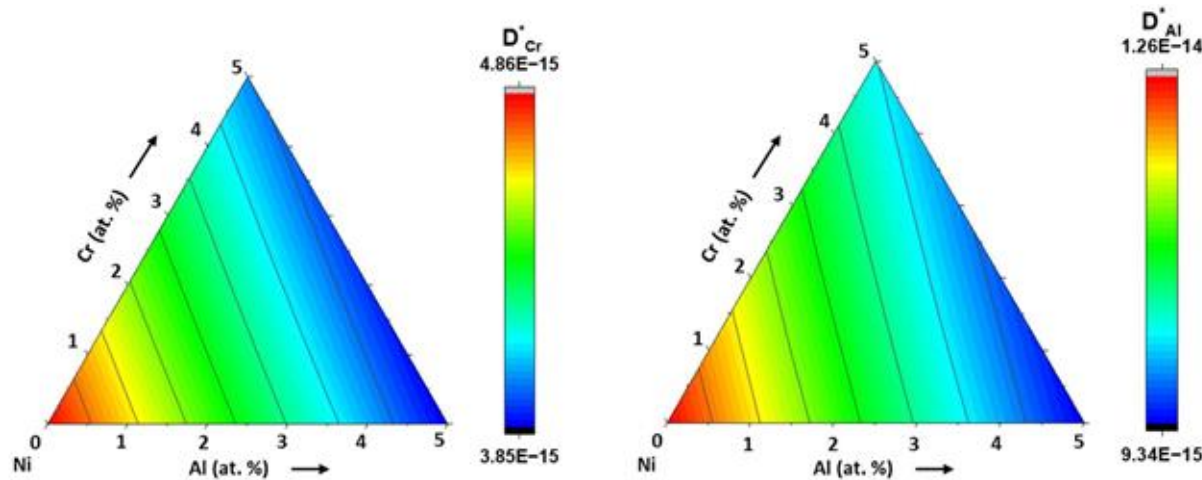
254 vs 253

Q(Al): activation energy, PINN vs experiment (kJ/mol)

Database-ready

parameter form is compatible with CALPHAD kinetic databases

The final result is a mobility map over composition



Tracer diffusion of Cr and Al over Ni–Cr–Al composition space at 1100 °C, inferred from two profiles

Ni–Co–Cr intrinsic fluxes obtained with a multi-objective evolutionary optimizer

The optimizer can change; physical constraints and independent anchors make the inversion informative

TAKEAWAYS

From OLS to a neural-network function

OLS and neural networks both fit parameters; the network supplies a much richer nested function family

Representation is not the governing law

A network may approximate Hall–Petch-like data, represent one PINN field, or represent a PINO map—the chosen object must be stated

Physics constrains inference; it does not remove ambiguity

Equations, boundary conditions, tracer anchors, and independent experiments are needed for physically useful diffusion parameters

One sentence to remember: always ask what function is represented, what information selects it, and how it is independently tested.

QUESTIONS & DISCUSSION

Questions

Saswata Bhattacharya

Department of Materials Science and Metallurgical Engineering
Indian Institute of Technology Hyderabad

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